Solid-Liquid Mixing in Stirred Vessels: Numerical Simulation, Experimental Validation and Suspension Quality Prediction Using Multivariate Regression and Machine Learning

Corentin MACQUERON
ORANO Projets
1 rue des hérons, 78180, Montigny-le-Bretonneux
corentin.macqueron@orano.group

Abstract
A two-phase solid-liquid stirred vessel modeling is performed with the FLUENT 15.0 software and validated by comparison to experimental data (power number and solid volume fraction profiles). The mixing quality, defined by the RSD (Relative Standard Deviation of the solid volume fraction), is studied as a function of the main system parameters: stirring speed, vessel diameter, pitch blade angle, liquid dynamic viscosity, and diameter, density and average volume fraction of the particles. On the basis of the ~300 numerical simulations performed with the experimentally validated model, a correlation to predict the RSD as a function of the different parameters with an accuracy of ±10 points of RSD (expressed in percentage) is proposed. Advanced machine learning techniques such as artificial neural networks can achieve an accuracy an order of magnitude better. Finally, it is shown that the correlation, built on a specific vessel and agitator design, can be applied as a pre-sizing to other designs with a reasonable accuracy, provided those designs are of the same type (newtonian axial downward pumping with D/T = ½ and presence of baffles).

Keywords
Solid-Liquid – Mixing – Suspension – Two-phase – Correlation – Machine Learning – Deep Learning – Neural Network – RSD – Euler Granular – Gidaspow – Brucato

1. Introduction

The solid-liquid mixing quality in a stirred tank as a function of the main system parameters (tank size and design, stirrer design, suspension characteristics) is an important matter for the industry. Correlations exist to predict the minimal stirring speed for bottom lifting or avoidance of settling (Zwietering, 1958; Mersmann *et al*., 1998) but these correlations are only meant for 'low' speeds where the mixing quality is generally poor. For 'high' speeds required to obtain a good mixing quality (defined by the RSD: Relative Standard Deviation of the solid volume fraction), there are only scarce elements in the literature and the construction of a general correlation is considered to be a difficult task (Kresta, 2016). To determine the stirring configuration allowing for a high suspension quality, experimentations or simulations are hence required. Real size experimentations are often not an option for different reasons, especially for very large tanks. Small scale experimentations can help, but scale-up laws are not always clearly established (Geisler *et al*., 1993; Montante *et al*., 2003; Dickey, 2013). Simulation is free from experimental constraints, but is still an onerous, difficult and cautious tool. In this context, ORANO (formerly AREVA) validated solid-liquid stirred tanks simulations performed with the FLUENT 15.0 software (ANSYS, 2013b) by comparison to experimental results produced by dedicated mock-ups in the framework of a preliminary study (Macqueron *et al*., 2015). In the present report, ORANO continues the validation task by comparison to experimental results from the literature (Tsz-Chung Mak, 1992) and present a numerical study to build a correlation aimed to predict the suspension quality as a function of the main system parameters. This correlation is based on ~300 numerical simulations ran on the Tsz-Chung Mak setup and is then tested against several other tank and impeller designs.

This technical report is an English follow-up of a previous study published in French in SFGP (Macqueron and William, 2017).

2. Experimental setup

The experimental setup chosen to validate the simulations and to build the correlation is the one from Tsz-Chung Mak because of its simplicity and because it is representative of many 'typical' industrial stirring systems. The main tank characteristics are the following:

- diameter: T = 0.61 m
- height: H = T
- clearance: C = T/4
- torispherical base (R = T, r = T/10)
- 4 baffles
- volume : 0.165 m^3

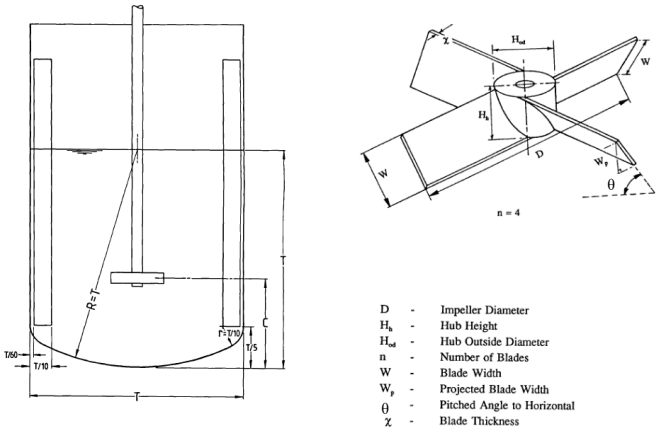


Figure 1. Tank and impeller schematic

The main impeller characteristics (4 pitch blades turbine at 45°, single stage) are shown in table 1 and figure 1. The power number Np is measured at 1.52 (with the mean suspension density).

Table 1. Impeller characteristics

D (m)	W (m)	χ (m)	H_b (m)	H_{od} (m)
3.10E-01	8.60E-02	3.40E-03	6.10E-02	5.70E-02

Tap water at ambient temperature is used as the liquid phase. The solid phase characteristics are the following (glass (sand) beads, sphericity: 0.8):

- density: 2600-2650 kg/m^3
- diameter: 150-210 µm (median value: 180 µm)
- concentration: 14% vol. (30% mass.)

The power is measured with a torque-meter (accuracy: ± 2%). The solid volume fraction φ is measured with a conductivity probe shown on figure 2. This probe measures the solid concentration between 5% and 50% vol. and is capable to detect a 1% variation.

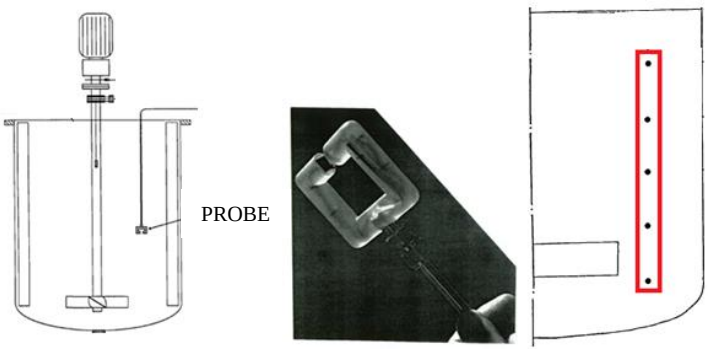


Figure 2. Conductivity probe at a T/6 distance to the wall and at T/6.1, T/3.1, T/2, T/1.5 and T/1.2 from the surface

The solid volume fraction is measured at the 5 locations shown in red on figure 2. The experimental RSD is calculated with these 5 measures according to the following formula (1):

$$RSD = \frac{1}{\varphi_{mean}}\left(\frac{1}{n-1}\sum_{i=1}^{n}(\varphi_i - \varphi_{mean})^2\right)^{\frac{1}{2}} \text{ (with n = 5)} \quad (1)$$

3. Modeling

3.1 Geometry and mesh

The geometry is built with the DesignModeler 15.0 software (ANSYS, 2013a) (figure 3). Because of the symmetries, only one quarter of the tank is modeled. The mesh (~800 k cells) is built with the ANSYS Meshing 15.0 software (ANSYS, 2013c) with boundary layers refinements on all the walls (5 layers, first layer thickness: 1 mm) to ensure y^+ (non-dimensional wall distance) values compatible with the selected turbulence model and wall functions (mean y^+ value: ~35, max y^+ value: ~130) (figure 3).

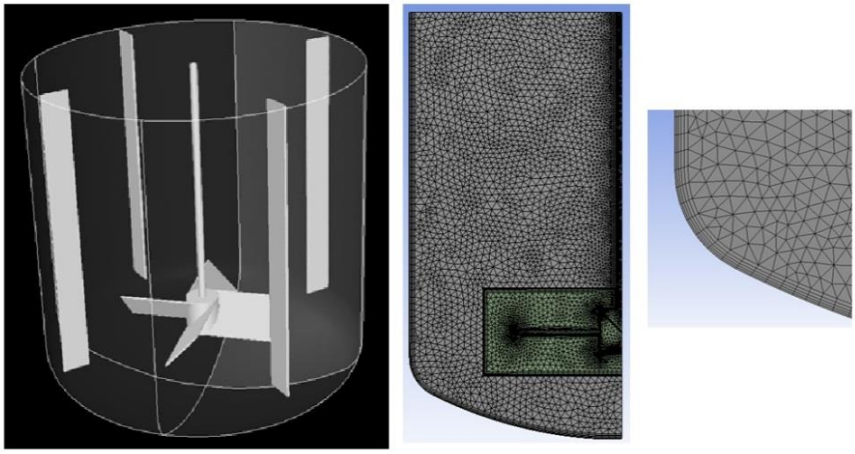


Figure 3. Tank geometry and mesh

3.2 Hypotheses, physical models and numerical parameters

The main hypotheses, physical models and numerical parameters are the following:

- the system is in steady-state
- the impeller rotation is modeled with the Multiple Reference Frame (MRF) method ('frozen rotor') (ANSYS, 2013b)
- turbulence is modeled with the *k-ε realizable mixture* approach with the *enhanced wall treatment* option (ANSYS, 2013b)
- the solid phase is modeled with the 'eulerian granular' approach, with the Syamlal-O'Brien, Lun *et al*. and Schaeffer models for granular viscosity (ANSYS, 2013b)
- the particles are perfectly spherical and their diameter is equal to the median value (180 µm)
- the particles density is equal to 2630 kg/m^3 as assumed by Tsz-Chung Mak
- the tap water viscosity is 0.001 Pa.s (newtonian fluid) and its density is 1000 kg/m^3 (values at ~20°C (Sacadura, 2009))
- the drag coefficient is modeled with the Gidaspow approach (Gidaspow, 1994, ANSYS, 2013b). No turbulent correction, such as the Brucato correction (Brucato *et al*., 1998), is taken into account (see § 4.2.7 for details). The Gidaspow approach is compared to other approaches in § 4.2.8
- lift is neglected
- the turbulent dispersion force (TDF) is modeled with the *Diffusion-in-VOF* approach (ANSYS, 2013b)
- the probe is not modeled (it is assumed to be non-intrusive)

- the free surface is a plane wall with a zero shear stress
- the pressure-velocity coupling is the *Coupled* approach (ANSYS, 2013b)
- the discretization scheme is *QUICK* (ANSYS, 2013b)
- the system is solved using the pseudo-transient approach with automatic time-step (ANSYS, 2013b)

4. Simulations

4.1 Validation

4.1.1 Power number

The power numbers determined with the simulations using the calculated impeller torque are given in table 2. The calculated values are very close to the measured values (maximal deviation is 2% taking into account the measurement accuracy). The simulations appeared validated for dissipated power prediction.

Table 2. Power number

N (RPM)	60	90	120	150	180	210	240	270	300
N_p (-)	1.51	1.51	1.52	1.53	1.56	1.58	1.58	1.58	1.58
Deviation from experiments	-1%	-1%	0%	1%	3%	4%	4%	4%	4%

4.1.2 Solid volume fraction

The solid volume fraction profiles from the simulations are compared to the measurements on figures 4 and 5. Some deviations are visible but the simulations appear to be capable to correctly reproduce the suspension variations as a function of the stirring speed, both qualitatively and quantitatively. The calculated RSD is very close to the measurements (maximal deviation is 5 points of RSD in %) as shown on figure 6. The numerical results are less conclusive at 'low' speed (60 RPM) (figure 4). At such 'low' speed, deactivating the turbulent dispersion force (TDF), as proposed by Tamburini *et al.* (2014), allows for a considerably better numerical prediction (figures 4 and 6). This 60 RPM speed being close and smaller to the avoidance of settling speed proposed by Mersmann *et al.* (N_{AS} (« Avoidance of Settling »), 72 RPM here), it seems that this N_{AS} speed could be seen as a 'speed criteria' under which the turbulent dispersion force should be deactivated. The 'low speed' regime thus appears different from the 'high speed' regime (Tamburini *et al.*, 2014). Further investigations should be performed but, from an engineering point of view, the simulation can be considered validated as long as one stays in the 'high speed' regime. The present report only aims at 'good quality' and 'high speed' predictions, so the turbulent dispersion force will always be activated in the following. An example of a solid volume fraction field obtained by the simulation is shown on figure 6.

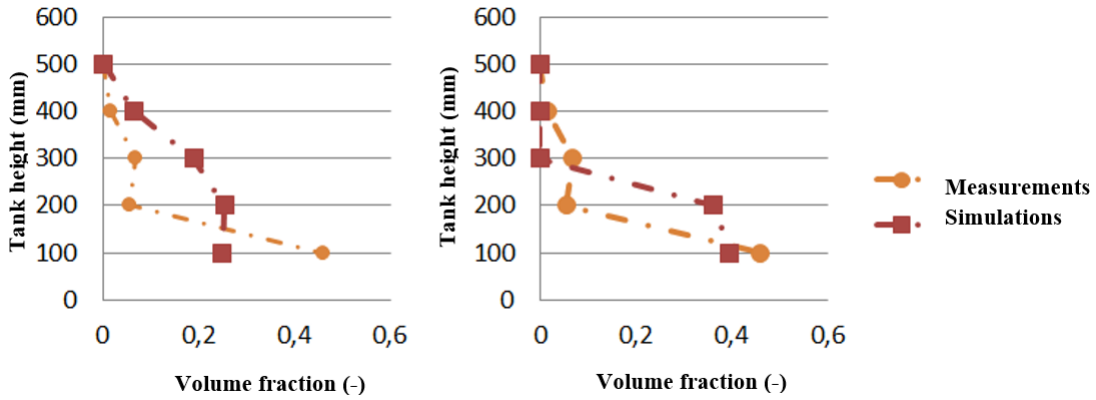


Figure 4. Simulated and measured solid volume fraction profiles – 60 RPM
(with turbulent dispersion force on the left, without turbulent dispersion force on the right)

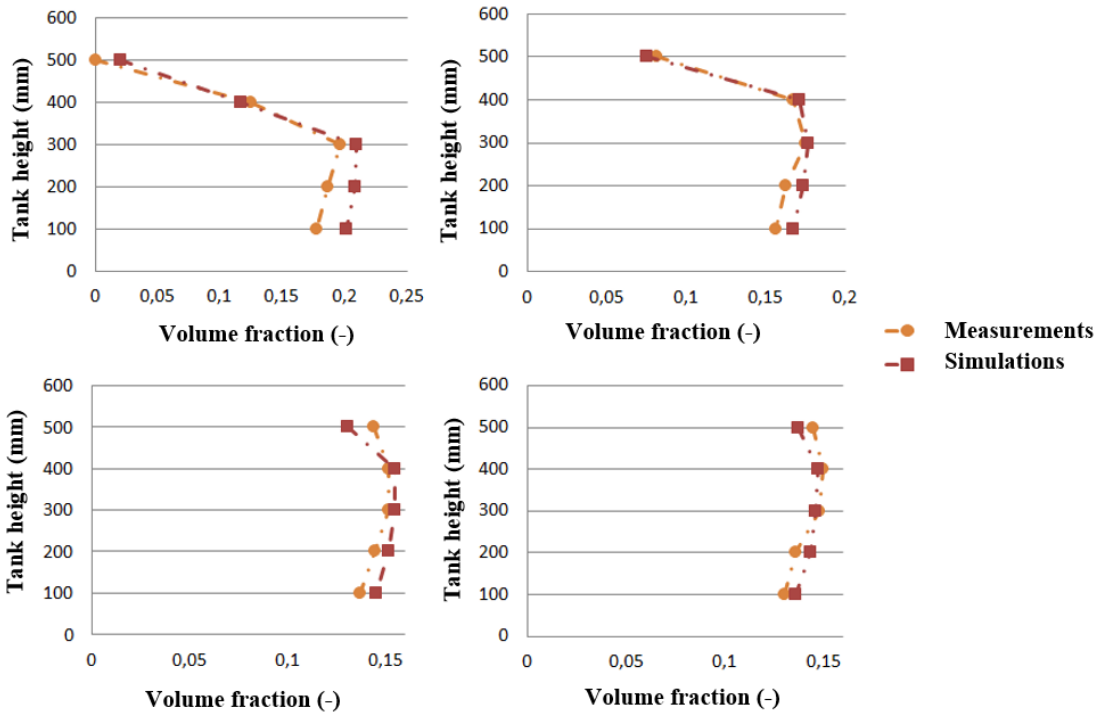


Figure 5. Simulated and measured solid volume fraction profiles.
Left to right, top to bottom: 90, 120, 150, 180 RPM

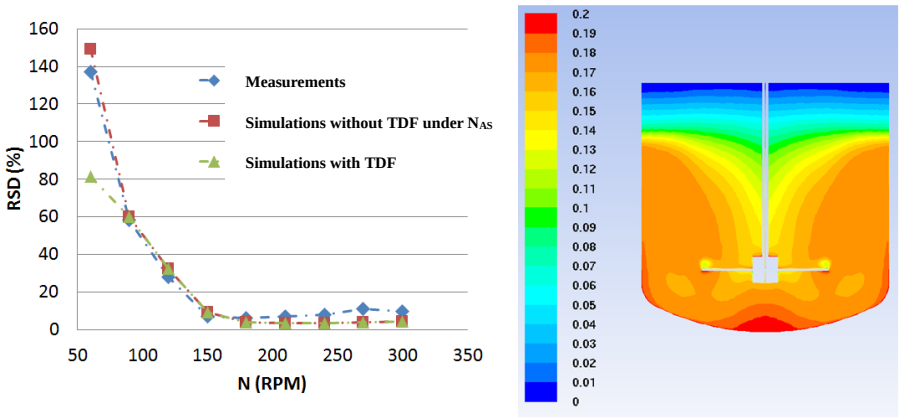


Figure 6. RSD versus stirring speed on the left (%).
Example of a solid volume fraction field (-) on the right (120 RPM)

4.2. Suspension quality

4.2.1. Tank diameter

Some results of the scale-up studies are shown on figure 7. The dependency in 0.814 exponent of the tank diameter is close to the one found by Zwietering (0.85).

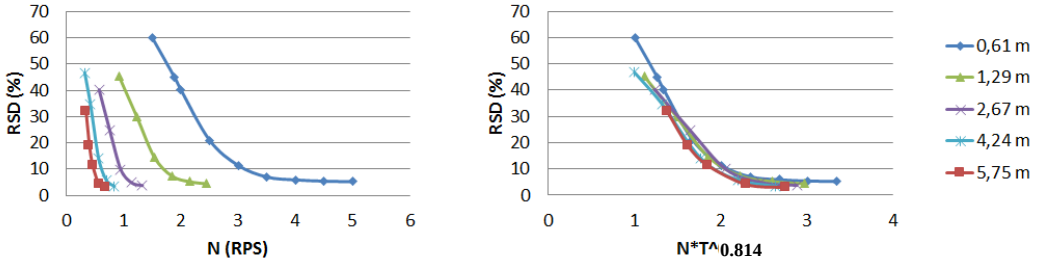


Figure 7. Tank diameter dependency

4.2.2 Particle diameter

Some results of the particle diameter studies are shown on figure 8. The dependency in -0.626 exponent of the particle diameter is notably different from the one of Zwietering (-0.2). This highlights the differences between the 'low' and 'high' speed physics.

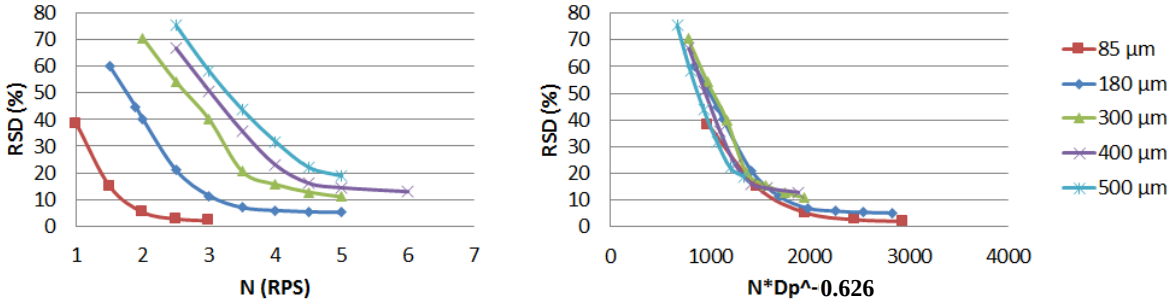


Figure 8. Particle diameter dependency

4.2.3. Particle density

Some results of the particle density studies are shown on figure 9. The results are shown as a function of the density difference over the liquid density, as in the Zwietering and Mersmann *et al.* formulae. The dependency in -0.445 exponent of the particle diameter is very close to the one of Zwietering (-0.45).

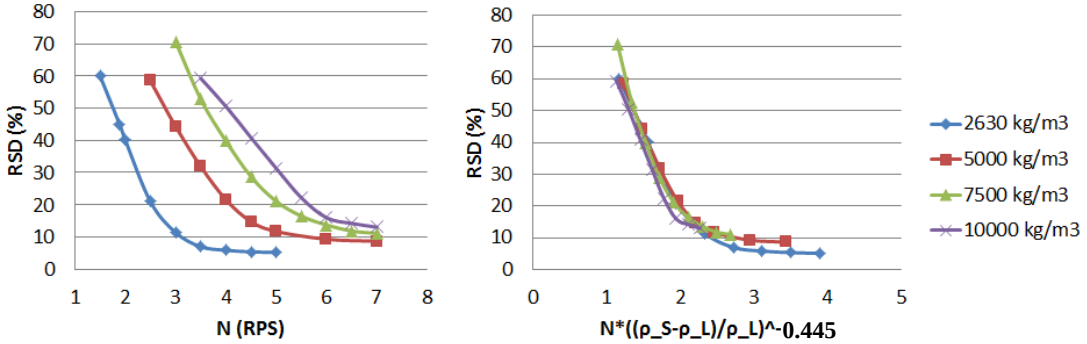


Figure 9. Particle density dependency

4.2.4. Liquid viscosity

Some results of the liquid dynamic viscosity studies are shown on figure 10. The dependency in 0.171 exponent of the particle diameter is notably different from the one of Zwietering (0.1). This highlights the differences between the 'low' and 'high' speed physics.

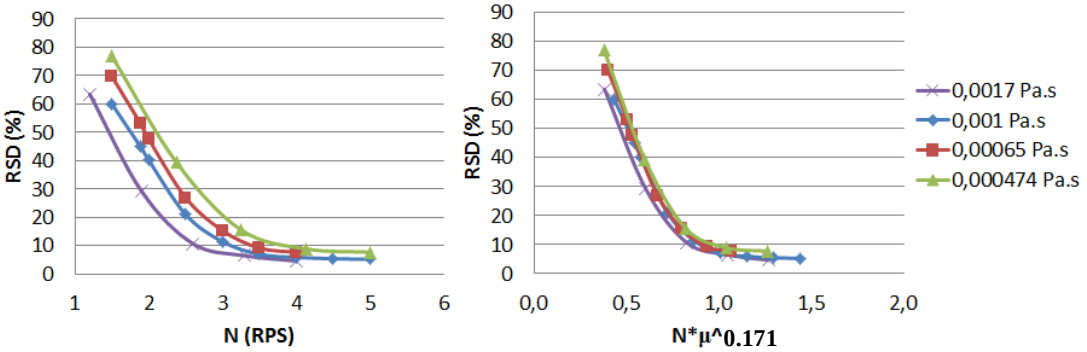


Figure 10. Liquid dynamic viscosity dependency

4.2.5. Solid volume fraction

Some results of the mean solid volume fraction studies are shown on figure 11. Figure 11 also shows that the RSD variation reaches a maximum ('bell curve'). The volume fraction for which this maximum is reached is somewhat variable but is always around 7.5%. This dependency is here approximated by an order 6 polynomial function (formula 11) and is very different from the one found by Zwietering which is in -0.13 exponent. This, again, highlights the differences between the 'low' and 'high' speed physics.

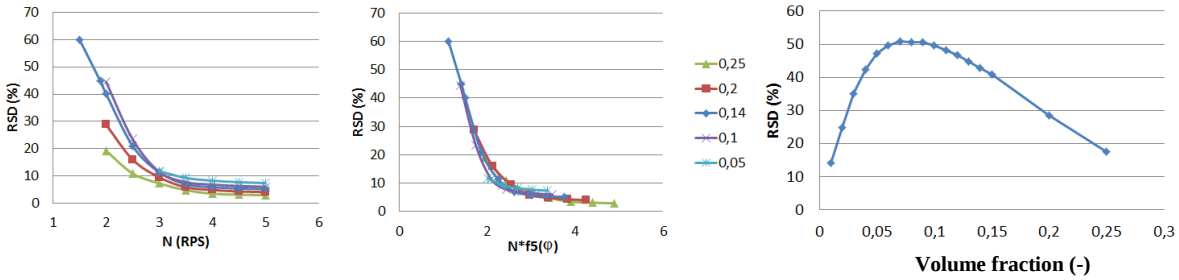


Figure 11. Mean solid volume fraction dependency

A possible explanation of this somewhat surprising non-monotonous dependency could be an antagonistic effect of the solid concentration. At 'low' concentrations (<~7.5%) the particles scarcely interact and the more particles there are, the more difficult it is to mix them. At 'high' concentrations (>~7.5%), the particles interact a lot because there are many of them, and these interactions increase the apparent fluid viscosity, which is beneficial to the mixing homogeneity (figure 11). Additional tests not presented here show that, above 7.5%, it is possible to correlate the RSD with the apparent viscosity calculated with the Krieger and Dougherty formula (1959), which is an indication that the two parameters may indeed be linked. It should nonetheless be noted that the terminal velocity of a particle in a still fluid is a strictly monotonous function of the volume fraction (Richardson and Zaki, 1954), but still sedimentation and stirred tanks are not the same. Further numerical and experimental investigations should be performed in order to better understand this behaviour and to confirm or disconfirm the 'bell curve'.

4.2.6. Pitch blade angle

The pitch blade angle results are shown on figure 12. The RSD dependency to the pitch blade angle being quite weak when compared to the other parameters studies in this report, it is neglected in the correlation detailed in § 5.1. The pitch blade angle has nonetheless a very important impact on the power consumption, as shown in table 3. There must be an optimum that should be looked at.

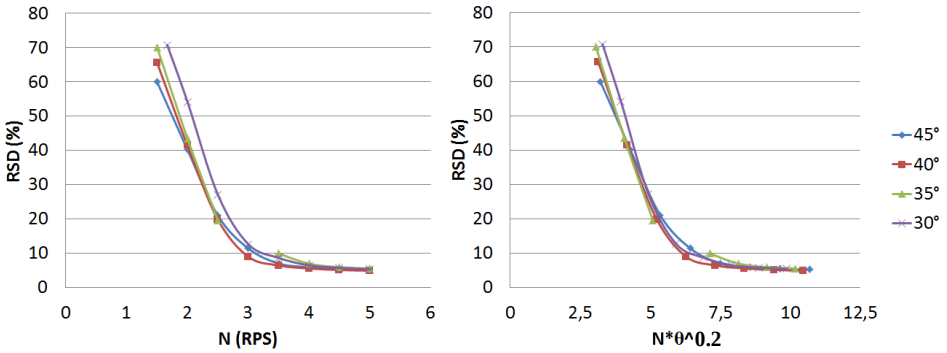


Figure 12. Pitch blade angle dependency

Table 3. Power number

θ (deg)	30	35	40	45
N_p (-)	~0.70	~0.95	~1.24	~1.56

4.2.7. Turbulence, drag coefficient and the Brucato correction

When the turbulence is strong, the drag coefficient on 'large' particles is increased as observed in several studies (Brucato *et al.*, 1998; Pinelli *et al.*, 2001; Montante *et al.*, 2004). The so-called 'Brucato' and 'Magelli' correlations have been proposed to modify the drag coefficient to account for this effect. These correlations appear to greatly improve the simulations accuracy in the aforementioned studies, whereas it seems to have some flaws in other configurations and appear to be non-universal. For instance, Kasat *et al.* (2008) used the Brucato correction but had to diminish its effect by a factor of ten to obtain correct results. We were also unable to completely reproduce the results from Montante *et al.* (2004) and Liu and Barigou (2013). In the latter case, we could accurately reproduce the experimental results for the 1 mm particles configurations for which the Brucato correction has little to no effects, but we were unable to reproduce the experimental results for the 3 mm particles configurations for which the Brucato correction has major effects: the correction seems to increase the drag coefficient by an over-estimated factor, resulting in over-suspended particles, especially in the lower part of the tank. Following the Kasat *et al.* approach (2008), we tried to diminish the value of the 'Brucato constant'. This approach had a great impact on the results, but no value of the Brucato constant was able to produce reasonable results (figure 13). In our previous study (Macqueron and William, 2017), we had activated the Brucato correction for all our calculations, but this had little to no effects in the tested configurations. In the present follow-up report, with larger particles (up to 1750 µm), the Brucato correction has an important impact. We decided to deactivate it, because, based on our observations, we were unsure of the correctness of its effects. Furthermore, it tends to over-suspend the particles and hence to over-estimate the mixing quality of the suspension, so we decided to stay on the safe side of the prediction. Further investigations should be performed in order to better understand this behaviour.

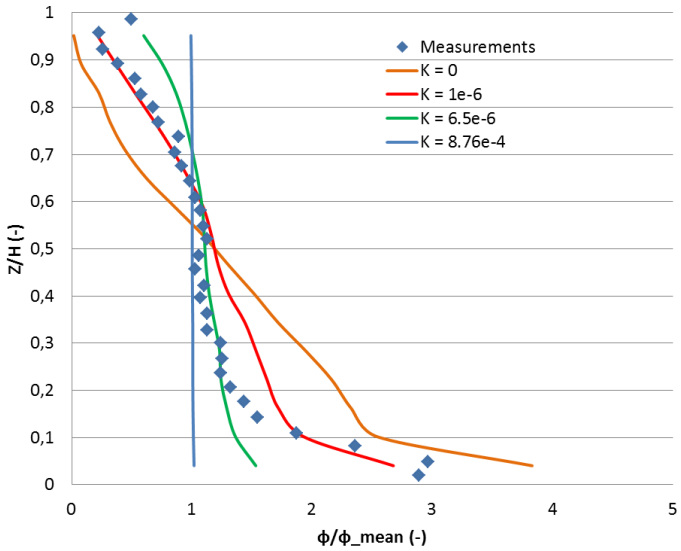


Figure 13. Brucato constant (K) dependency

4.2.8. The Gidaspow drag correlation

It is sometimes said that the Euler-Granular approach is not significantly better than the simpler Euler-Euler approach. Our sensitivity studies indeed showed that it is rather the drag coefficient correlation that is of crucial importance to accurately mimic the physics of the solid suspension. The Gidaspow drag correlation proved to be significantly better than other approaches (figure 14) and, as it is only available in the Euler-Granular approach in FLUENT 15.0, we had to use the Euler-Granular approach even if it was not for the Granular approach *per se*.

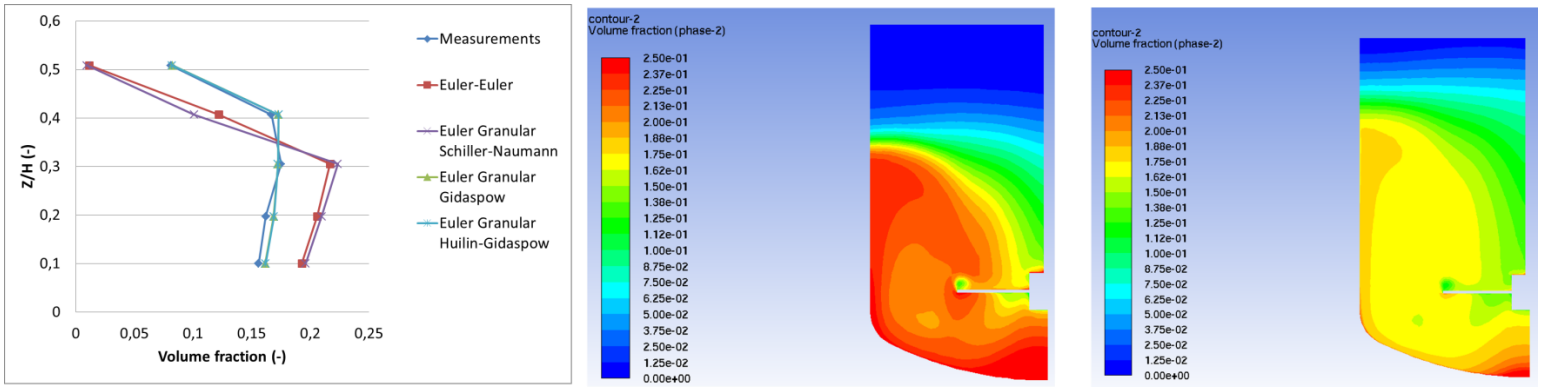


Figure 14. Drag coefficient sensitivity studies.
(Left: volume fraction profiles for different drag approaches, center: volume fraction field for Euler-Granular Schiller-Naumann, right: volume fraction field for Euler-Granular Gidaspow)

5. Prediction

The mixing quality (RSD) has been studied by performing ~300 numerical simulations with the validated model. Some of these simulations are shown in § 4 and the total span of the study is the following:

- Stirring speed (N), from ~N_{AS} to ~5 N_{AS}
- Tank diameter (T), from 0.61 m to 6.50 m
- Pitch blade angle (θ), from 30° to 45°
- Particles diameter (D_p), from 40 µm to 1750 µm
- Solid-liquid relative density ($\Delta\rho/\rho$), from 0.74 (ρ_S = 1740 kg/m^3 (magnesium)) to 17.95 (ρ_S = 18 950 kg/m^3 (uranium))
- Mean solid volume fraction (φ), from ~0.5% to 30%
- Liquid dynamic viscosity (µ), from 4.74.10^{-4} Pa.s to 1.7.10^{-3} Pa.s (newtonian fluid)

From now on, the RSD is calculated on the entire volume of the tank with the following formula (2):

$$RSD = \frac{1}{\varphi_{mean}} \sqrt{\frac{1}{V} \iiint (\varphi(V) - \varphi_{mean})^2 dV} \quad (2)$$

Among the ~300 calculations, roughly half has been performed with the configuration described in § 2 by modifying only one parameter at a time to help visualize the variations, shown in § 4. The other half has been performed by modifying several or all the parameters at a time in order to maximize the size of the studied domain and to take into account the possible interactions between the parameters.

5.1 Multivariate regression

With the ~300 numerical simulation results, a correlation (formula 3) has been built by multivariate regression (least square error) in order to predict the mixing quality. It appears that the multivariate dependency of the RSD can be approximated with a reasonable accuracy by the product of monovariate functions (formulae 6 to 12):

$$RSD = \alpha e^{\gamma \beta^{\omega}} \text{ (in \%)} \quad (3)$$

With:

$$\alpha = 144{,}022 \quad (4)$$

$$\gamma = -3.36 \times 10^{-3} \quad (5)$$

$$\omega = 1.364 \quad (6)$$

$$\beta = N \times f_1(T) \times f_2(D_p) \times f_3\left(\frac{\Delta\rho}{\rho}\right) \times f_4(\mu) \times f_5(\varphi) \quad (7)$$

$$f_1(T) = T^{0.840} \quad (8)$$

$$f_2(D_p) = {D_p}^{-0.702} \quad (9)$$

$$f_3\left(\frac{\Delta\rho}{\rho}\right) = \left(\frac{\Delta\rho}{\rho}\right)^{-0.449} \quad (10)$$

$$f_4(\mu) = \mu^{0.203} \quad (11)$$

$$f_5(\varphi) = 69574.98\varphi^6 - 68045.96\varphi^5 + 26995.72\varphi^4 - 5516.34\varphi^3 + 610.77\varphi^2 - 33.42\varphi + 1.36 \quad (12)$$

Figure 15 shows the RSD values from the simulations as a function of the β variable.

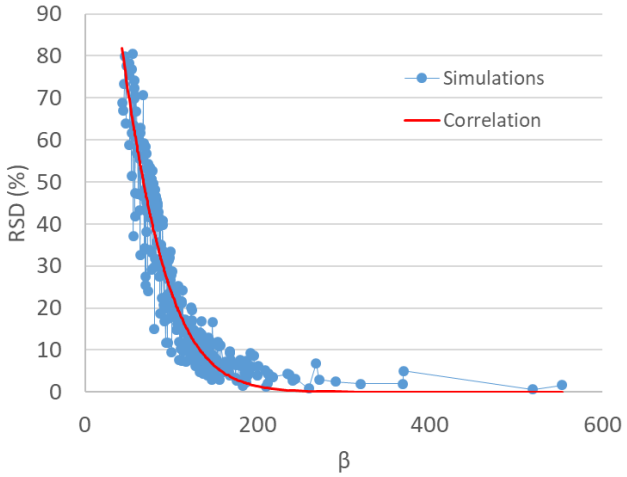


Figure 15. Comparison between correlation predictions and 3D simulations

This correlation has been built on the Tsz-Chung Mak design only. Eventhough this design is a very common one and makes the correlation widely applicable, its validity for other tank and impeller designs is necessarily brought into question. In order to assess our correlation robustness, it was tested against several other tank and impeller designs, shown on figure 16. All these designs have in common baffles, a D/T = ½ ratio and a downward pumping axial flow, but they are different in terms of overall designs, power numbers, pitch blade angle, blade numbers and H/T ratios. In these ~50 additional configurations, the tank diameter is comprised between 0.69 m and 6.5 m, the particle diameter is comprised between 40 µm and 1000 µm, the mean solid volume is comprised between ~0.5% and 12%, the particle density is comprised between 2500 kg/m^3 and 4500 kg/m^3, the liquid density is comprised between 992 kg/m^3 and 1150 kg/m^3 and the liquid dynamic viscosity is comprised between 6.5x10^{-4} Pa.s and 1.1x10^{-3} Pa.s.

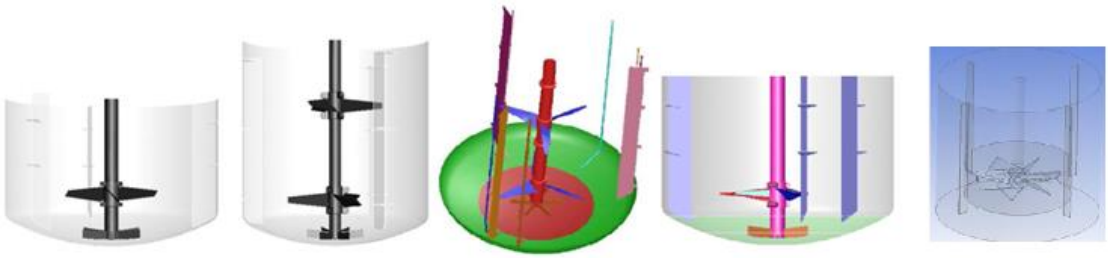


Figure 16. Designs from left to right: HRB-1 – HRB-2 – RCB – HRB-3 – LIU
Power numbers from left to right: 0.62 – 1.27 – 0.28 – 0.18 – 1.6
H/T from left to right: 0.68 – 1.1 – 1.23 – 0.68 – 1

The numerical results obtained in these configurations are plotted against the β variable defined in § 5.1, without any additional fitting (figure 17). The results are somewhat dispersed but are still in very good agreement with our proposed correlation which can hence be considered validated for pre-sizing mixing calculations for any newtonian downward pumping axial systems with D/T = ½ and baffles. Lower and upper estimation curves are provided.

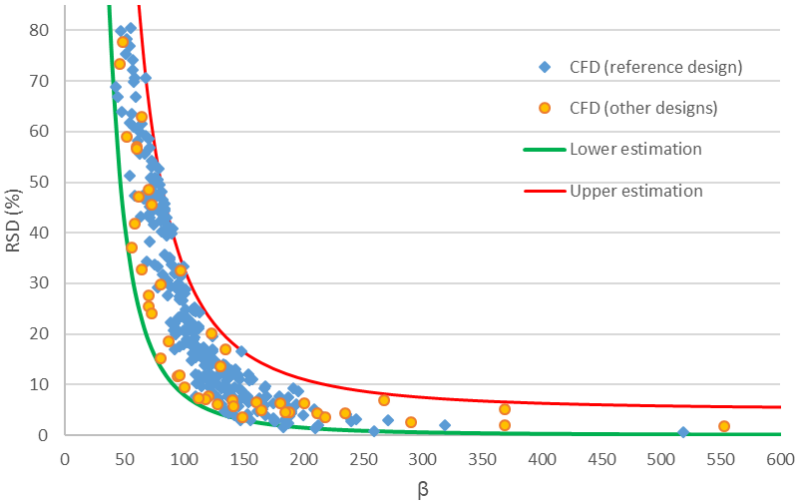


Figure 17. Validation of the correlation for several tank and impeller designs

$$Lower\ estimation = 500000 \times \beta^{-2,4}$$

$$Upper\ estimation = 700000 \times \beta^{-2,2} + 5$$

5.2 Machine Learning

After the multivariate regression presented in § 5.1, we decided to go further by using machine learning techniques. (Technically, the aforementioned regression already fall in the 'machine learning' domain, in its broader sense, but we prefer to use the term 'machine learning' for more advanced techniques.)

We hence built machine learning regressors such as k-Nearest-Neighbors, Support Vector Machine, Decision Tree, Random Forest, Bagging, AdaBoost, Gradient Boosting, eXtreme Gradient Boosting and Gaussian Process regressors using *Scikit Learn* (Pedregosa *et al.*, 2011). We also built artificial neural networks which appeared to be especially efficient using *Keras* (Chollet, 2015) and *Deep Float* (Macqueron and Fragnaud, 2017).
Training, hyperparameters tuning and overfitting avoidance was performed by splitting the data into three sets: a training set (75% of the data) to 'teach' the models, a validation set (25% of the data) for blind test and hyperparameters tuning and a final set (5% of the data) for final model performance-overfitting evaluation. For neural networks, early stopping combined with callback of the best fit is a major tool to avoid overfitting.

The hyperparameters of the models were chosen by performing a parameter grid search and by choosing the model with the lowest mean squared error on the validation set among the models having a validation/training error ratio below a threshold equal to 5. The validation set was never given to the models during the learning process, but as the hyperparameters were chosen to have the lowest mean squared error on the validation set, some form of indirect and implicit 'data leaking' is introduced into the process and the models could perform unreasonably well on the validation set and still generalize poorly on completely unseen data. The final set, never seen by the models during both training and hypeparameters tuning phases, ensures the generalization capabilities of the models.

In total, more than 200 000 machine learning models were tested. Figures 18 to 29 show the best model for each type of regressor (regressor predictions vs FLUENT results, error histogram with gaussian fitting showing the mean error (μ), the mean absolute error (MAE) and the standard deviation of the error (σ)). Each model is compared to the linear model which serves as the reference model.

Linear

```
RMSE = 0.079683
RMSE Training = 0.077782
RMSE validation = 0.084912
Ratio Validation/Training = 0.916022
```

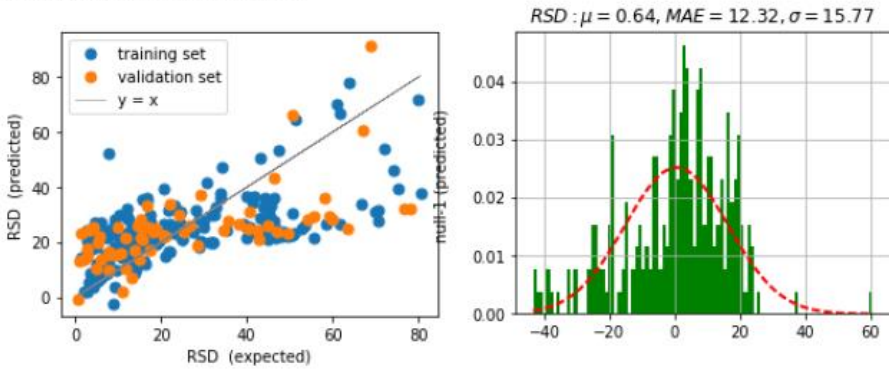


Figure 18. Linear regressor

Polynomial (10 AIs tested) 6th order

```
RMSE = 0.071170
RMSE Training = 0.071643
RMSE validation = 0.069792
Ratio Validation/Training = 1.026521
```

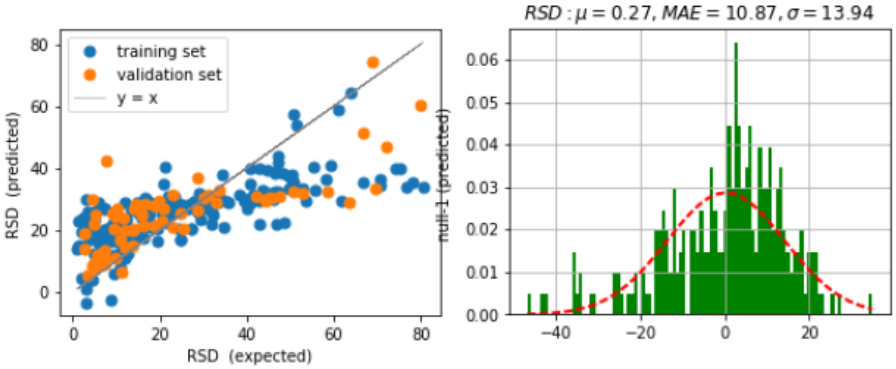


Figure 19. Polynomial regressor

Support Vector Machine (3 AIs tested)

```
RMSE = 0.081481
RMSE Training = 0.081423
RMSE Validation = 0.081648
RMSE Final = 0.074602
Ratio Validation/Training = 0.997244
Ratio Final/Validation = 1.094454
```

Kernel : Radial Basis Function

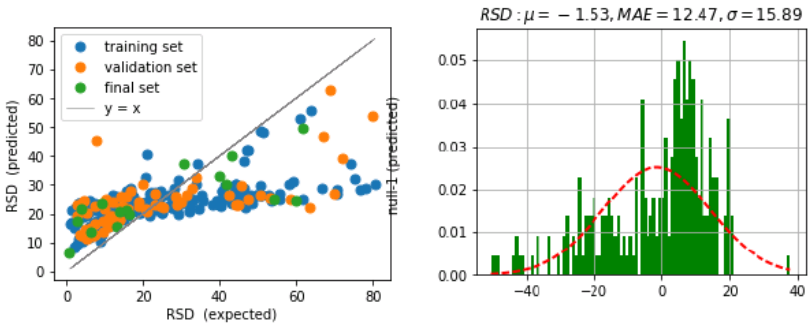


Figure 20. Support Vector Machine regressor

kNN (4488 AIs tested)

```
RMSE = 0.042271
RMSE Training = 0.035825
RMSE Validation = 0.056870
RMSE Final = 0.056195
Ratio Validation/Training = 0.629944
Ratio Final/Validation = 1.012023
```

k : 2
Leaf size : 1
Weights : uniform
Algorithm : auto

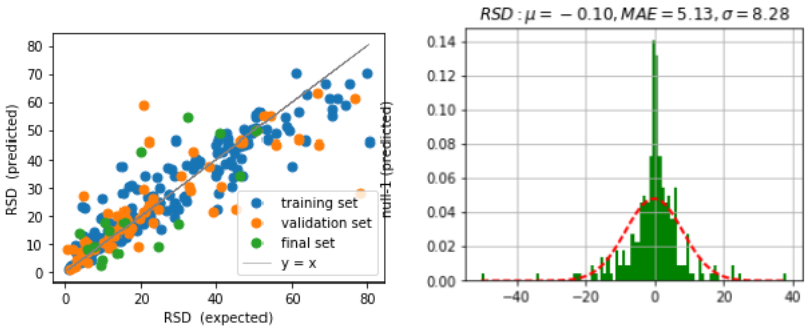


Figure 21. k-Nearest-Neighbors regressor

Gaussian Process (4176 AIs tested)

```
RMSE = 0.013959
RMSE Training = 0.007990
RMSE Validation = 0.023906
RMSE Final = 0.020405
Ratio Validation/Training = 0.334238
Ratio Final/Validation = 0.841609
```

Alpha : 0.0001
Kernel : Rational Quadratic
Length scale : 1
Normalize : False

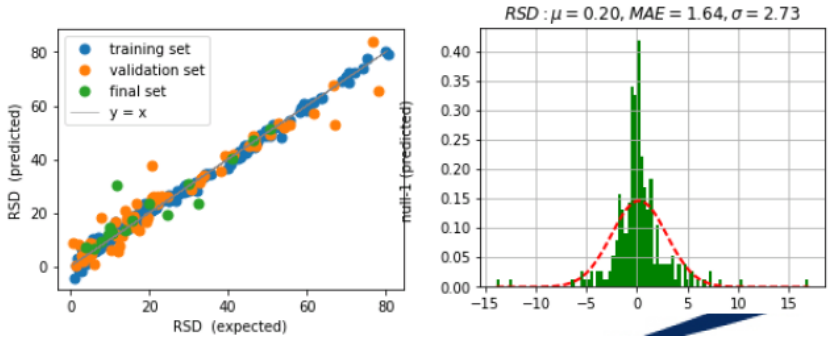


Figure 22. Gaussian Process regressor

Decision Tree (3840 AIs tested)

```
RMSE = 0.036905
RMSE Training = 0.031151
RMSE Validation = 0.049880
RMSE Final = 0.043473
Ratio Validation/Training = 0.624517
Ratio Final/Validation = 1.147382
```

Max depth : 200
Max features : auto
Min samples leaf : 2

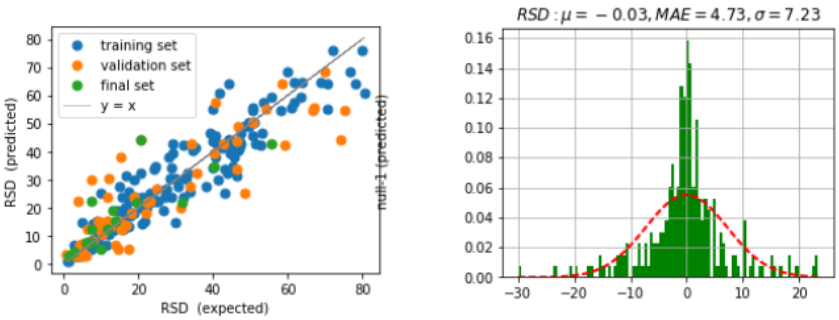


Figure 23. Decision Tree regressor

Random Forest (1440 AIs tested)

```
RMSE = 0.022846
RMSE Training = 0.019260
RMSE Validation = 0.030921
RMSE Final = 0.033706
Ratio Validation/Training = 0.622861
Ratio Final/Validation = 0.917386
```

N_estimators : 16
Max depth : none
Max features : auto
Min samples leaf : 1

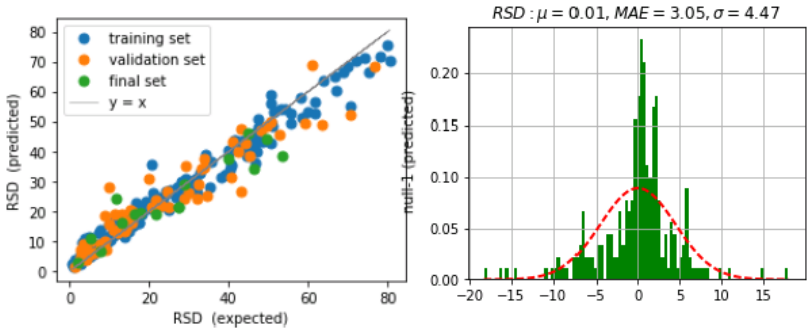


Figure 24. Random Forest regressor

Bagging (1440 AIs tested)

```
RMSE = 0.089473
RMSE Training = 0.091932
RMSE Validation = 0.081994
RMSE Final = 0.101363
Ratio Validation/Training = 1.121204
Ratio Final/Validation = 0.808915
```

Base estimator : decision tree
N_estimators : 5
Max features : 4
Max samples : 4

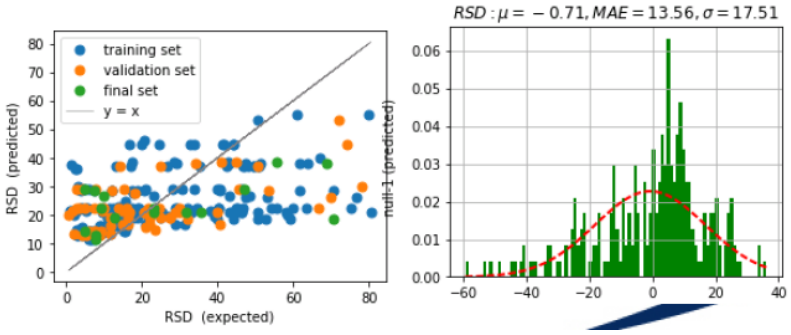


Figure 25. Bagging regressor

AdaBoost (384 AIs tested)

```
RMSE = 0.063337
RMSE Training = 0.061301
RMSE Validation = 0.068857
RMSE Final = 0.064643
Ratio Validation/Training = 0.890264
Ratio Final/Validation = 1.065188
```

Base estimator : decision tree
Learning rate : 1
N_estimators : 750

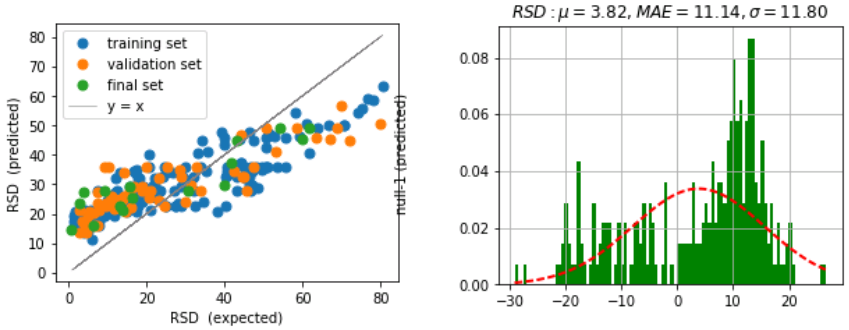


Figure 26. AdaBoost regressor

Gradient Boosting (98 280 AIs tested)

```
RMSE = 0.022927
RMSE Training = 0.014314
RMSE Validation = 0.038050
RMSE Final = 0.034565
Ratio Validation/Training = 0.376181
Ratio Final/Validation = 1.100838
```

N_estimators : 1024
Min sample leaf : 5
Max features : auto
Max depth : none
Learning rate : 0.01

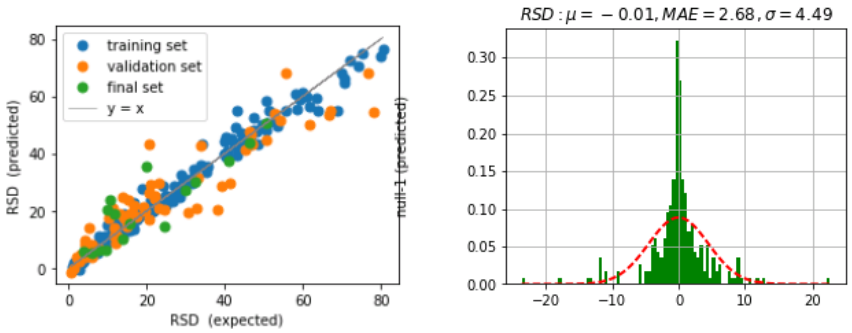


Figure 27. Gradient Boosting regressor

eXtreme Gradient Boosting (98 280 AIs tested)

```
RMSE = 0.024106
RMSE Training = 0.018541
RMSE Validation = 0.035545
RMSE Final = 0.032894
Ratio Validation/Training = 0.521620
Ratio Final/Validation = 1.080574
```

N_estimators : 2048
Min samples leaf : 50
Max features : auto
Max depth : 3
Learning rate : 0.01

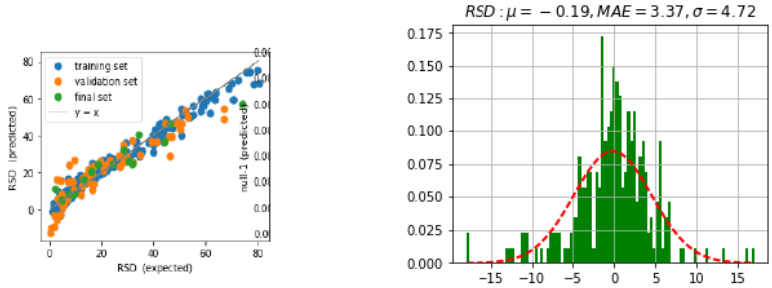


Figure 28. eXtreme Gradient Boosting regressor

Neural Network (990 AIs tested)

```
RMSE = 0.006401
RMSE Training = 0.004676
RMSE Validation = 0.009792
RMSE Final = 0.013070
Ratio Validation/Training = 0.477569
Ratio Final/Validation = 0.749205
```

Activation : ReLU
Optimizer : ADAM
Hidden layer : 1
Neurons per layer : 512
Dropout : 0%
Epoch max : 1000
(early stopping + callback)

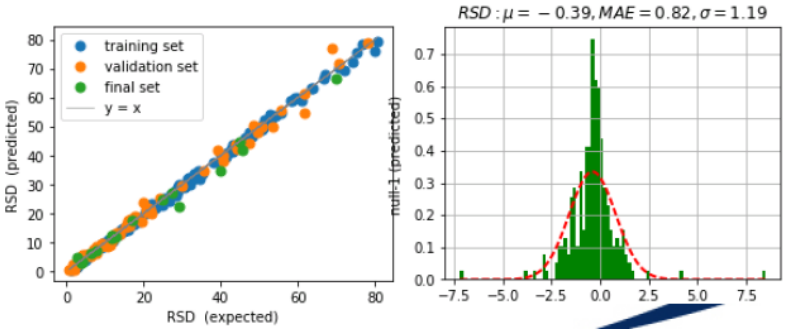


Figure 29. Neural Network regressor

Figure 30 shows the performance of each model. It is clear that the neural network approach outperforms every other machine learning regressor in this study (its accuracy is an order of magnitude better than the correlation's one). Having more neurons in the hidden layer (512) than the number of cases (~300) might seem specious and prone to overfitting, but the very good performance on both validation and final sets proves otherwise. Here, overfitting is avoided by early stopping during training and callback of the optimal fit.

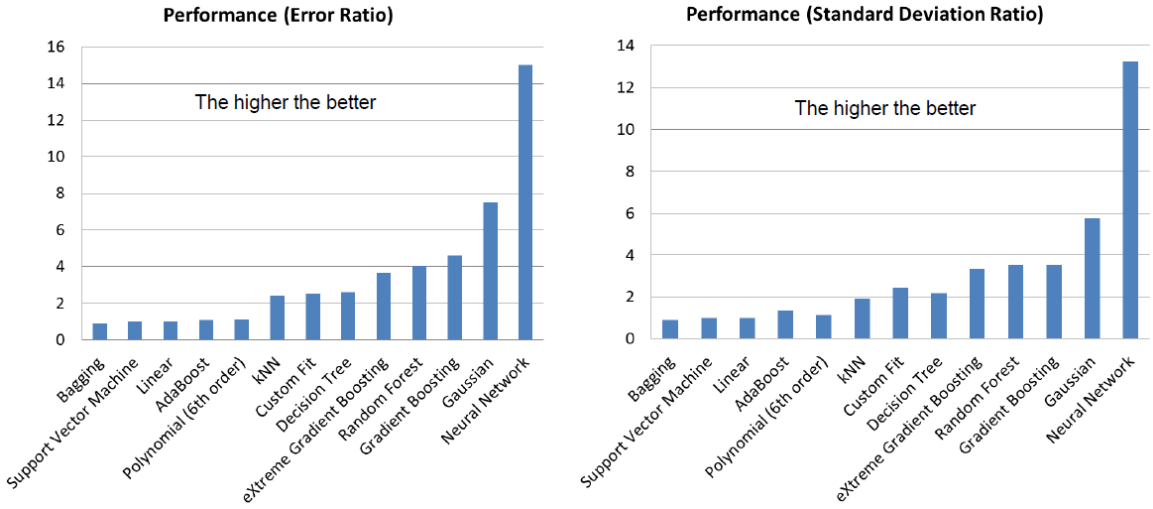


Figure 30. Regressor performance (comparison to the linear model which serves as the reference model) ('custom fit': our correlation (§ 5.1))

The neural network generalization capabilities to other designs is however poorer than the correlation, probably because, being able to learn much more accuracy and being only trained on one design, the neural network becomes much more design-dependent than the correlation. When trained with the data from all the designs, its mean accuracy deteriorates but its generalization capabilities increase and finally outperform the correlation (figure 31).

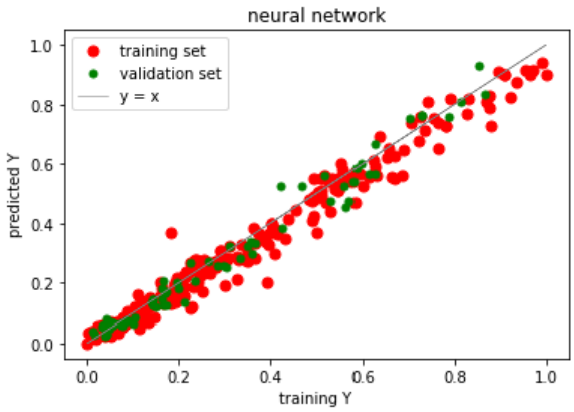


Figure 31. Neural network performance on all the designs (scaled) (final set not shown here but comparable to validation set)

Neural networks are prone to behave erratically outside of the domain they were trained on (figure 32 (Lohninger, 1999)). We hence tested our neural network inside and outside of its training domain to look for possible unrealistic behaviour. Figure 33 shows the behaviour of the model around its convex hull training domain center, variable by variable. Negative values are not possible and some behaviours are questionable, but at least the neural network appears capable to keep a consistent behaviour for most

variables for both interpolation and extrapolation. Other regressors show clear signs of inconsistency (even inside their training domain) (figure 34).

Whereas it would be wrong to draw any general conclusion, it is clear that, in our case, the neural network approach completely outperforms the other tested machine learning techniques in terms of mean, max and standard deviation error and interpolation-extrapolation behaviour.

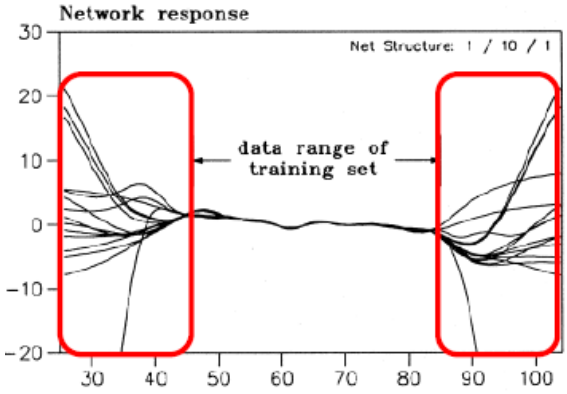


Figure 32. An example of a neural network erratic behaviour in extrapolation (Lohninger, 1999)

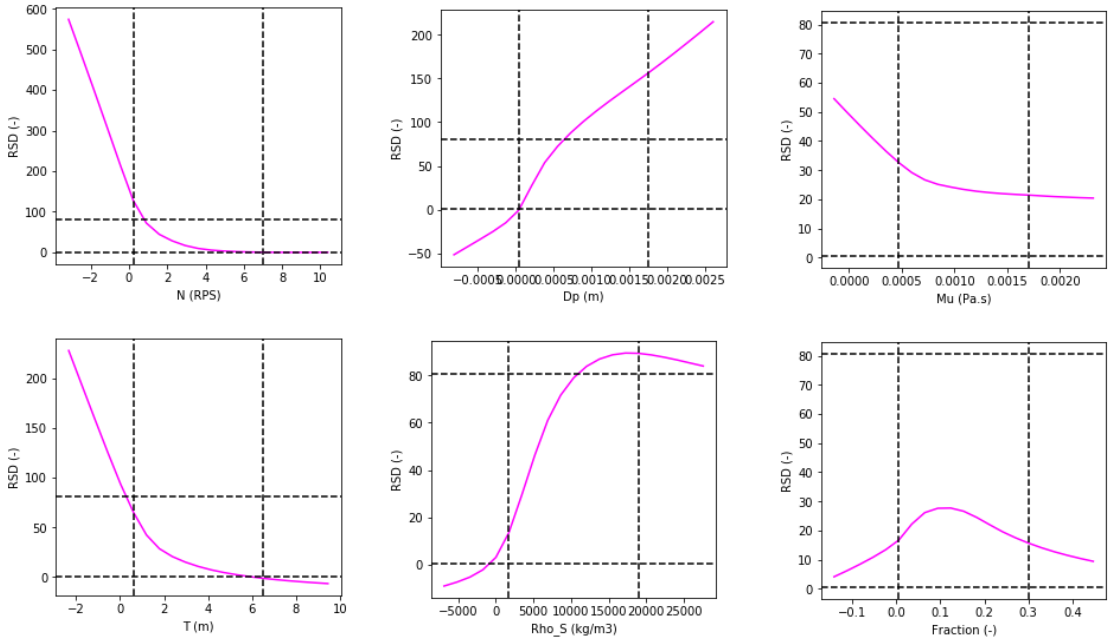


Figure 33. Neural network behaviour inside and outside its training domain (inside: center square)

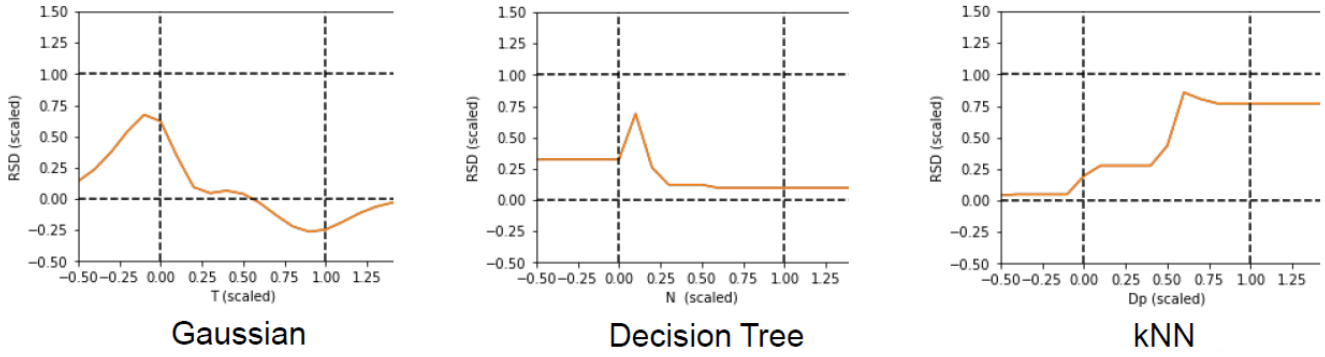


Figure 34. Various regressors behaviours inside and outside their training domain (inside: center square)

6. Conclusion

It is shown in this report that FLUENT 15.0 solid-liquid simulations in stirred tanks can accurately reproduce dissipated power and solid volume fraction profiles when the stirring speed is above the 'avoidance of settling' speed as defined by Mersmann *et al*. (1998). Under this criterion, the simulations deteriorates but can still produce good results by deactivating the turbulent dispersion force, as also shown by Tamburini *et al*. (2014).

The mixing quality of the suspension was studied as a function of the main system parameters: stirring speed, tank diameter, pitch blade angle, liquid viscosity, particles diameter, particles density and mean solid volume fraction (~300 simulations with a validated model). Based on these results, a correlation predicting the suspension quality defined by the RSD (Relative Standard Deviation, expressed in %) as a function of the aforementioned parameters was built. It is also shown that this correlation, built for a particular yet common tank and impeller design can produce relevant results for pre-sizing with other designs, provided they are of the same type (downward pumping, newtonian liquid, with baffles and D/T = ½ ratio). With advanced machine learning techniques such as neural networks the accuracy of the prediction can be even better.

7. Perspectives

Further numerical and experimental investigations should be performed in order to better understand the transition between 'low' and 'high' speed regimes, the effect of the turbulence on the drag coefficient ('Brucato correction'), the influence of the mean solid volume fraction ('bell curve') and to optimize the tank and impeller design in order to obtain the best suspension quality with the minimal dissipated power.

Nomenclature

ADAM	ADAptive Momentum	
AI	Artificial Intelligence	
TDF	Turbulent Dispersion Force	
QUICK	Quadratic Upstream Interpolation for Convective Kinematics	
ReLU	Rectified Linear Unit	
RSD	Relative Standard Deviation	(-)
C	clearance	(m)
D	impeller diameter	(m)
D_p	particle diameter	(m)
H	tank height	(m)
H_b	hub height	(m)
H_{od}	hub diameter	(m)
MAE	Mean Absolute Error	(-)
N	stirring speed	(RPM, RPS)
N_{AS}	avoidance of settling stirring speed	(RPM, RPS)
N_p	power number	(-)
R, r	curvature radius	(m)
T	tank diameter	(m)
W	blade width	(m)
y^+	non-dimensional wall distance	(-)
α, β, γ	correlation coefficients	
φ	solid volume fraction	(-)
χ	blade thickness	(m)
μ	dynamic viscosity or mean error	(Pa.s) or (-)
ρ_S	particle density	(kg/m^3)
ρ_L	fluid density	(kg/m^3)
σ	standard deviation of the error	(-)
θ	pitch blade angle	(°)

References

Brucato, A., Grisafi, F., Montante, G. (1998). Particle drag coefficients in turbulent fluids. *Chemical Engineering Science*, vol. 53, n° 18, pp. 3295-3314.

Chollet, F. (2015). Keras: Deep Learning Library for Theano and TensorFlow. https://keras.io/

DesignModeler 15.0. (2013). Canonsburg: ANSYS.

Dickey, D. S. (2013). Ten Things You May Not Know About Liquid Mixing Scaleup. *Chemical Engineering*. New York: McGraw Hill.

Fluent 15.0. (2013). Canonsburg: ANSYS.

Gidaspow, D. (1994). *Multiphase Flow and Fluidization: Continuum and Kinetic Theory Description*. Boston: Academic Press.

Geisler, R. K., Buurman, C., Mersmann, A. (1993). Scale-up of the necessary power input in stirred vessels with suspension, *The Chemical Engineering Journal*, vol. 51.

Hicks, T. G., Chopey, N. P. (2012). *Handbook of Chemical Engineering Calculations*. McGraw Hill.

Kasat, G. R., Khopkar, A. R., Ranade, V. V., Pandit, A. B. (2008). CFD simulation of liquid-phase mixing in solid-liquid stirred reactor, *Chemical Engineering Science*, vol. 63, pp. 3877-3885.

Kresta, S. M., Etchells III, A. W., Dickey, D. S., Atiemo-Obeng, V. A. (2016). *Advances in Industrial Mixing. A Companion to the Handbook of Industrial Mixing*. Hoboken: Wiley. p. 386.

Krieger, I. M., Dougherty, T. J. (1959). A mechanism for non-newtonian flow in suspensions of rigid spheres. *Transaction of the Society of Rheology*, vol. 3, pp.137-152.

Liu, L., Barigou, M. (2013). Numerical modelling of velocity field and phase distribution in dense monodisperse solid-liquid suspensions under different regimes of agitation: CFD and PEPT experiments. *Chemical Engineering Science*.

Lohninger, H. (1999). *Teach/Me Data Analysis*. Berlin-New York-Tokyo: Springer-Verlag. ISBN 3-540-14743-8

Macqueron, C., William, J., Le Gall, R., Demarthon, R., Piot, G., Ragouilliaux, A. (2015). Waste homogenisation tank: experimental validation of a numerical study. In: *Proceedings of Global*. Paris: SFEN, paper 5204.

Macqueron, C., William, J. (2017). Suspension diphasique liquide-solide en cuve agitée: une corrélation de prédiction de la qualité du mélange sur la base de simulations numériques validées sur mesures expérimentales. *Récents Progrès en Génie des Procédés*, n° 110, ISSN: 1775-335X, ISBN: 978-2-910239-85-5. Paris: SFGP.

Macqueron, C., Fragnaud, S. (2017). *Deep Float: An Easy Deep Learning Tool in Your Internet Browser*. Research Gate. https://deep-float.herokuapp.com

Mersmann, A., Werner, F., Maurer, S., Bartosch, K. (1998). Theoretical prediction of the minimum stirrer speed in mechanically agitated suspensions. *Chemical Engineering and Processing*, vol. 37 (issue 6), pp. 503-510.

Meshing 15.0. (2013). Canonsburg: ANSYS.

Montante, G., Pinelli, D., Magelli, F. (2003). Scale-Up Criteria for the Solids Distribution in Slurry Reactors Stirred with Multiple Impellers. *Chemical Engineering Science*, vol. 58, pp. 5363-5372.

Montante, G., Bakker, A. (2004). Solid-Liquid Multiphase Flow Validation in Tall Stirred Vessels with Multiple Impeller Systems. Fluent Technical Note 253.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, vol. 12, pp. 2825-2830. https://scikit-learn.org/stable/

Pinelli, D., Nocentini, M., Magelli, F. (2001). Solids distribution in stirred slurry reactors: influence of some mixer configurations and limits to the applicability of a simple model for predictions. *Chem. Eng. Comm.*, 118, pp. 91-107.

Richardson, J. F., Zaki, W. N. (1954). Sedimentation and fluidisation: part 1, *Transactions of the Institution of Chemical Engineers*, vol. 32, pp. 35-41

Sacadura, J. F. (2009). *Initiation aux transferts thermiques*. 8ème édition. Paris: Tec&Doc, p. 430.

Tamburini, A., Brucato, A., Cipollina, A., Micale, G., Ciofalo, M. (2012). CFD Predictions of Sufficient Suspension Conditions in Solid-Liquid Agitated Tanks. *International Journal of Nonlinear Sciences and Numerical Simulation*, vol. 13 (issue 6), pp. 427-443.

Tamburini, A., Cipollina, A., Micale, G., Brucato, A., Ciofalo, M. (2014). Influence of drag and turbulence modelling on CFD predictions of solid liquid suspensions in stirred vessels. *Chemical Engineering Research and Design*, vol. 92 (issue 6), pp. 1045-1063.

Tsz-Chung Mak, A. (1992). *Solid-liquid mixing in mechanically agitated vessels*. PhD. University of London.

Zwietering, T. N. (1958). Suspending of solid particles in liquid by agitators. *Chemical Engineering Science*, vol. 8 (Issues 3-4), pp. 244-253.